# Quantum Entanglement

Leonard Susskind

Supplementary track, 2006 Fall Part 1
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

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# Chapter 1

# Quantum Entanglements, Part 1

These notes follow Leonard Susskind's opening lecture on quantum entanglement, with curation by LazyingArt LLC. The lecture begins, characteristically, with a sermon about intuition. It then turns that sermon into a program: we will not begin with wave mechanics, but with information, bits, configurations, update rules, and only then the elementary algebra that makes those update rules precise. The chapter should therefore be read in that order. We are being taught not merely a list of facts, but a way of rebuilding intuition.

## 1.1 Why Quantum Mechanics Feels Unnatural

Susskind begins by reminding us that our ordinary physical intuition is an evolutionary inheritance. It was formed in a world of rocks, arrows, prey, predators, and moderate velocities. Even an animal chasing another animal is already solving a problem in kinematics: it tracks relative velocity, direction, and the moment at which the sign of that relative velocity changes. So one should not underestimate how much classical mechanics is already wired into ordinary behavior.

But modern physics begins precisely where that inheritance stops being reliable. Relativity concerns velocities that no human or animal ever encountered in ordinary life. Quantum mechanics concerns electrons, uncertainty, and the breakdown of the very trajectory-based picture that ordinary experience encourages us to adopt. A particle, according to common sense, is a thing that has a position at every instant; if it has a position at every instant, then it has a trajectory; if it has a trajectory, then we compute its velocity from the differences of positions. That whole chain of intuition comes from the world of thrown stones and flying arrows. It is not a guide to the microscopic world.

So the problem is not that quantum mechanics is illogical. The problem is that it lies outside the regime that trained our brains. We therefore need abstraction before we can recover intuition. Susskind says, in effect, that physicists rewire themselves. They learn abstract mathematics first, and only afterward develop new habits of thought that make the subject feel less alien.

That sermon is not decorative. It explains the method of the lecture. This will not be a conventional first course in quantum mechanics, centered from the outset on the Schrödinger equation and the wave picture. The emphasis will be different. We are going to concentrate on the logic of quantum mechanics and the logic of quantum information. That is why the lecture pivots almost immediately to its first concrete object: the bit.

## 1.2 Classical Bits and the Counting of States

Physics, in this register, is information. To say something about a physical system is to provide information about it, often numerically, but very often in the form of a discrete alternative. Heads or tails, yes or no, up or down: these are all examples of the same logical structure.

**Definition 1.1.** A bit is a question about a system with exactly two possible answers.

Susskind insists on starting with the classical case. A classical bit may be represented by 0 and 1, although the choice of symbols is arbitrary. We could equally well use 1 and $-1$, or any other pair of labels. The point is only that there are two possibilities. Once that is fixed, we can write the two classical states as

$$|0\rangle, \qquad |1\rangle. \tag{1.1}$$

At this stage the Dirac notation looks like excess baggage, and Susskind says so. It adds nothing to the classical description *yet*. But it is installed here because it will be the natural notation later, when the bit becomes quantum.

For many bits we simply write a string inside the ket:

$$|001011\cdots\rangle. \tag{1.2}$$

This is a configuration of a multibit system. The notation is intentionally plain. We are not yet taking tensor products apart; we are just recording a string of binary values.

Now comes the first quantitative question: how many states does such a system have? One bit has two states. Two bits have four. Each new bit doubles the number of possibilities. Therefore an $n$-bit classical system has

$$N_s = 2^n \tag{1.3}$$

possible configurations, where $N_s$ is the number of states. Inverting the relation gives

$$n = \log_2 N_s. \tag{1.4}$$

This is the point where the lecture turns the word *bit* from a label into a measure. The number of bits of information carried by a system is, by definition, the logarithm base two of the number of available states:

$$I = \log_2 N_s. \tag{1.5}$$

For systems built from binary alternatives this reproduces an integer. But the definition is broader. A die has six states, so its information content is $\log_2 6$, which is not an integer. The definition survives even when the system is not literally built from a set of binary switches.

This is a small but important beat in the lecture. Susskind first defines the bit operationally, then counts states, then inverts the relation. The movement is deliberate: we are being taught to think of a state space as something whose size already measures information.

## 1.3 How Ordinary Physics Becomes a Bit String

The next question is the natural one: are bits merely a convenient language for toy systems such as coins, or do they really reach ordinary physics? Susskind's answer is emphatic. Almost any physical

information can be represented by bits, at least approximately and often to arbitrarily improving approximation.

Begin with a single measured quantity, such as the temperature of a room. If we care only about finite accuracy, then we may truncate the value and write it in base two. Once that is done, the number is simply a finite string of zeros and ones. If we want more accuracy, we keep more places. If we want a finer notion of temperature, we redefine the unit and again represent the result in binary. The essential move is the same every time: finite accuracy turns a real-valued quantity into a long but finite bit string.

The same idea applies to fields. A field is something that varies from place to place. To encode it, we divide space into many small cells, order the cells, and record the field value in each cell with some chosen precision. The result is a long sequence of binary strings, one block per cell, and therefore a long sequence of bits. In that sense, a temperature field, an electric field, or a magnetic field may all be represented by binary data once we agree to discretize space and tolerate finite precision.

Susskind then gives a second example, even more directly in the language of yes-or-no questions. Impose a lattice on a room and suppose that no cell can contain more than one particle. Then each cell carries one bit:

$$\text{occupied} \longleftrightarrow 1, \qquad \text{empty} \longleftrightarrow 0. \tag{1.6}$$

A particle configuration is thus encoded by a string of occupancies. What does the string mean? It means exactly this: the first cell is occupied, the second is empty, the third is empty, the fourth is occupied, and so on. We have traded a spatial picture for a sequence of answers to yes-or-no questions.

This is one of the lecture's recurring motifs: *so far no quantum mechanics.* Nothing quantum has entered yet. We are still talking about classical information, classical bits, and classical approximation schemes. But the point is already substantial. If physical information can be encoded by bit strings, then digital computation is not an accident. That is why computers can simulate physics at all.

There is, however, a caveat, and the lecture states it explicitly. Exact classical real numbers may require infinitely many bits. Rational numbers may repeat, irrational numbers certainly do not terminate, and long-time classical tracking may demand more and more precision. But that does not disturb the practical or conceptual point being made here. It only sharpens it: a computer works because finite approximation is often enough, and because physical descriptions can be improved systematically by using more bits and finer discretization.

## 1.4 Configurations, Update Rules, and State Space

Up to this point the lecture has concentrated on how to *describe* a system at one time. The next pivot is from description to law. A physical theory is not only a list of possible configurations; it must also tell us how one configuration becomes another.

**Definition 1.2.** A configuration is the state of a system at a given instant of time.

Susskind is careful here. First comes configuration, not motion. We specify the state at one instant. Then, if we wish, we may discretize time as well as space and speak of updating from one digital

instant to the next. In that language a physical theory consists of two parts: a space of possible states and a rule that updates one state into another slightly later state.

To make this visible, the lecture introduces an abstract state space. Its points are not points in ordinary space; they are entire configurations. That distinction matters. A closed loop in this state space is not a particle running around a spatial circle. It is a system moving through a cycle of possible configurations.

For a one-bit system the state space has only two points. Then the simplest possible update laws can be written in bare form:

$$0 \mapsto 0, \qquad 1 \mapsto 1, \tag{1.7}$$

or

$$0 \mapsto 1, \qquad 1 \mapsto 0. \tag{1.8}$$

The first is the trivial fixed-point law: nothing happens. The second is the flip-flop law: the system alternates forever between the two possibilities.

The same idea extends immediately to two bits. A two-bit system has four states. One perfectly good law moves the system around a closed loop in the logical space of those four configurations. Another good law describes two separate flip-flops happening in parallel. The lecture is not trying to exhaust the possibilities. It is trying to install a picture: the laws of motion may be drawn as arrows between states.

That is why the section matters. We are being led from a string of bits to a geometry of possibilities, and from there to dynamics as a pattern of arrows. Once that picture is in place, one further question becomes unavoidable: are all arrow patterns equally good?

## 1.5 Reversibility, Information Loss, and Unitarity

Now the lecture reaches its conceptual center. Susskind first shows examples of acceptable update laws, and then he draws another law that is logically easy to imagine but physically defective. Two distinct states flow into a single later state. Going forward is still defined. Going backward is not.

The good diagrams all have the same structural feature. Each state has one definite next state, but also one definite previous state. Susskind phrases this as the *uniqueness of the future* and the *uniqueness of the past*. That is what reversibility means here.

### 1.5.1 Question & Answer

**Question.** Why is the merging update rule forbidden? It seems perfectly sensible as a forward rule. If the system is here, it goes there. Why is that not enough?

**Answer.** Because the rule destroys information about the past. If two distinct starting points both flow into the same later point, then once we arrive there we can no longer tell where we came from. The future is defined, but the past is not uniquely reconstructible. In Susskind's language, the rule loses information. At the fundamental level, where we imagine keeping track of every degree of freedom and not merely a coarse-grained summary, classical physics does not allow that.

This is the rhythm of the lecture: first a picture that looks reasonable, then the recognition that something has gone wrong, and then the sharpened principle that repairs it. The repair is not

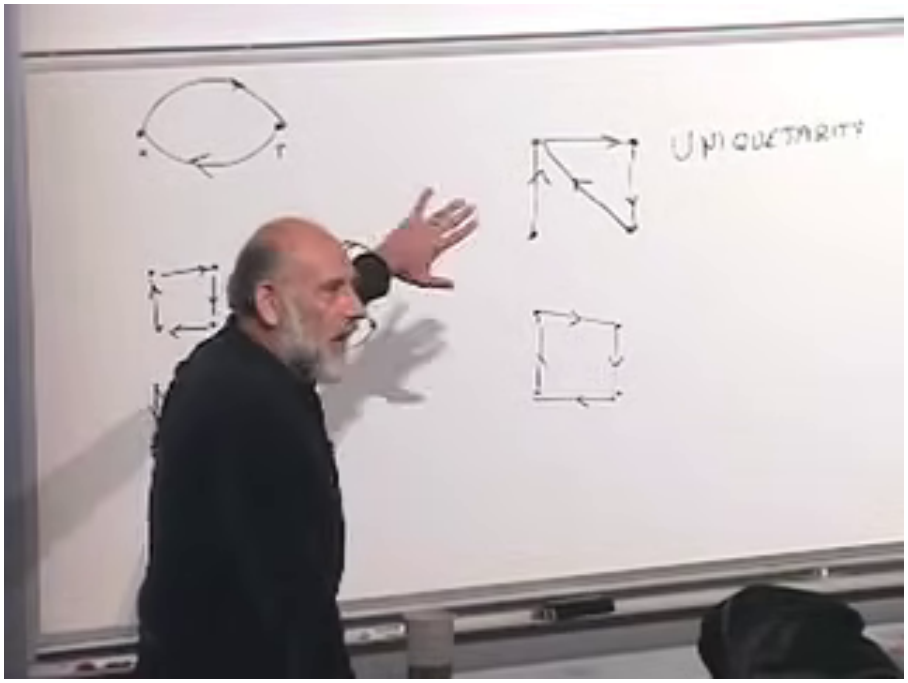



Figure 1.1: Board evidence from the discussion of reversible and irreversible update rules. The handwritten word *unitarity* appears beside the emphasized four-state sketch.

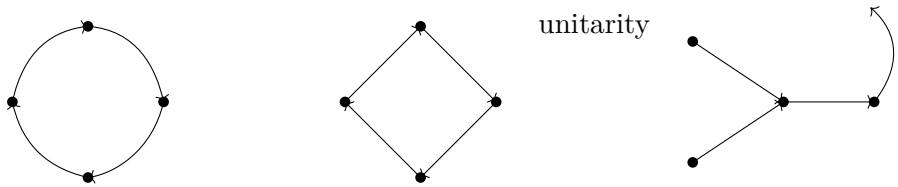



Figure 1.2: A clean schematic reconstruction of the board logic: reversible loops on the left, and a contrasting information-losing update rule on the right. The redraw stays schematic because the board does not securely label the states.

a matter of logic alone. Susskind explicitly says that such information-losing rules are logically conceivable; they are simply not the form taken by fundamental physical laws.

A second clarification follows immediately. Reversibility is not quite the same thing as exact time-reversal symmetry. A cyclic law still has an orientation: under forward evolution it runs one way around the loop, not both ways at once. What reversibility means is that there exists a reverse law taking us back uniquely. So the lecture draws a distinction between *time reversibility* and the stronger notion of exact symmetry under interchange of past and future.

A third clarification comes from thermodynamics. It often *looks* as if information is lost. Release the gas from one corner of a room and after a while the whole room looks the same no matter which corner we started from. But, Susskind argues, this is usually not fundamental loss. It is coarse-graining. We have stopped keeping track of the microscopic details. If we followed every molecule with enough precision, then in principle the information would still be there. What is lost in practice is our power to resolve it.

This is also the place where the lecture gestures beyond classical physics. Susskind jokes about

naming the property "unique-tarity," and then gives the quantum name: *unitarity*. The quantum theory will have its own formal statement of this idea, but the principle has already appeared in classical dress: past and future must both be unique.

## 1.6 Vectors, Rows, Columns, and Inner Products

After the break Susskind says, in effect, that he was about to jump into quantum mechanics, but he wants to stop first and lay out some elementary mathematics. The timing matters. The algebra is not an isolated appendix; it is being installed because the classical discussion has made it necessary.

A vector is introduced in the least ceremonial possible way. Susskind does not begin with axioms of a vector space. He says plainly: a vector is a finite sequence of numbers. At this point the entries are real numbers. Complex conjugation, daggers, and the full quantum notation are postponed.

A four-component vector may be written as a row,

$$A = (A_1, A_2, A_3, A_4), \tag{1.9}$$

or as a column,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}. \tag{1.10}$$

The two forms contain the same information. They are simply different notations adapted to different operations. Susskind even warns us not to visualize these as arrows in ordinary space. For the present purpose they are just organized lists of numbers.

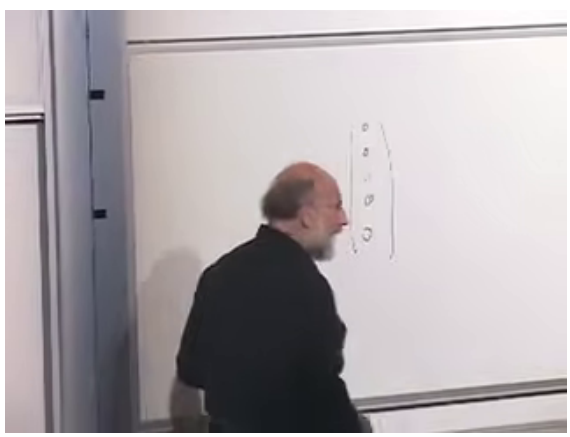


Figure 1.3: The board at the moment the lecture isolates a column vector. The entries are too small to transcribe literally, so the clean reconstruction uses generic components $A_1, \ldots, A_4$.

Now take a second vector, written as a row:

$$B = (B_1, B_2, B_3, B_4). \tag{1.11}$$

If the row and the column have the same number of entries, we may multiply them. The rule is the familiar one:

$$BA = B_1A_1 + B_2A_2 + B_3A_3 + B_4A_4. \tag{1.12}$$

This is the inner product. It is not another vector. It is a single number. In ordinary three-dimensional language it would be called the dot product, and Susskind says so, but he prefers the more general name because the objects are being treated abstractly.

The pedagogical point is simple and operational. Rows multiply columns to produce scalars. Once that is secure, matrices can be introduced with almost no extra conceptual burden.

## 1.7 Matrices as Update Operators

A matrix is now presented as a machine that acts on a vector and produces a new vector. That description is intentionally concrete. We are not yet being handed the formal theory of linear operators. We are being given the piece of algebra that will encode update laws.

A $4 \times 4$ matrix is written as

$$M = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix}. \tag{1.13}$$

Its rows and columns are themselves collections of numbers, but together they define one operation. Acting on the column vector $A$, the matrix produces a new column vector whose entries are row-by-column inner products:

$$(MA)_i = \sum_j M_{ij} A_j. \tag{1.14}$$

Written out componentwise, the first two entries are

$$(MA)_1 = M_{11}A_1 + M_{12}A_2 + M_{13}A_3 + M_{14}A_4, \tag{1.15}$$

$$(MA)_2 = M_{21}A_1 + M_{22}A_2 + M_{23}A_3 + M_{24}A_4. \tag{1.16}$$

The later entries are obtained in exactly the same way. Susskind does two and leaves the pattern to the audience.

**A worked classical example.** Now the lecture turns back to the classical discussion and shows why matrices matter. Suppose the system has finitely many configurations arranged in a cycle. Susskind first tries to draw a five-state example, then abandons it and retreats to four states so that the arithmetic will fit. That change does not alter the mathematical point.

We encode the four classical states by one-hot column vectors:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.17}$$

A cycle $1 \to 2 \to 3 \to 4 \to 1$ may then be represented by the permutation matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.18}$$

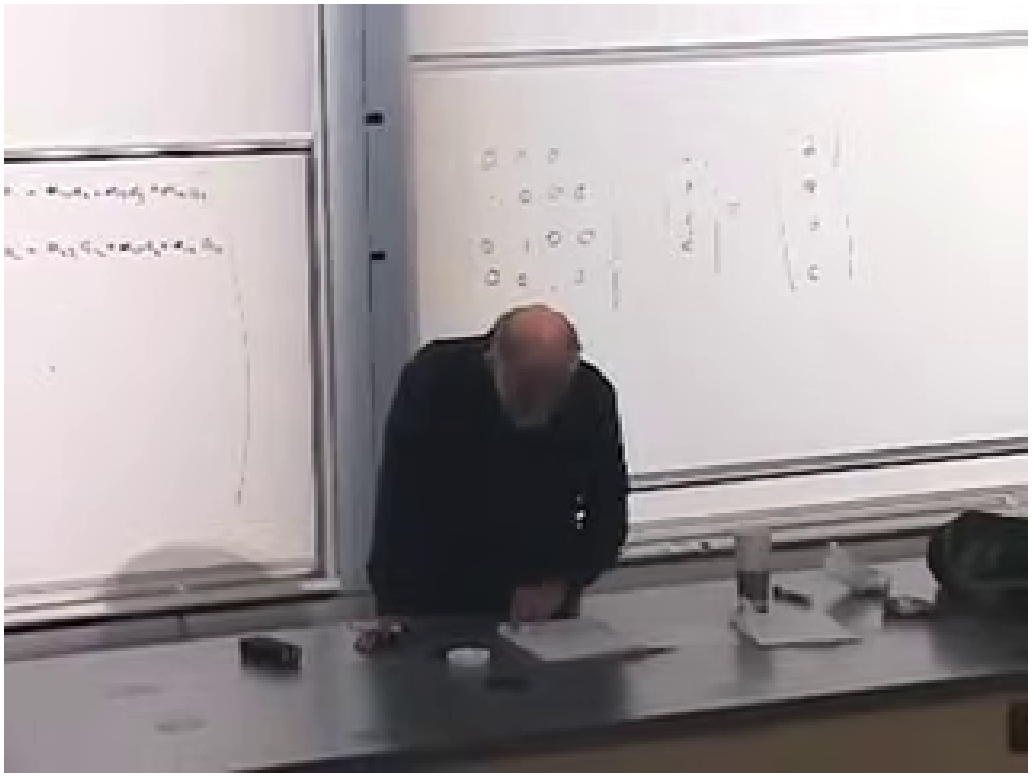


Figure 1.4: Board layout for the discussion of a matrix acting on a column vector. The frame preserves the spatial organization of the explanation more reliably than the individual micro-entries.

This is a deliberate clean convention. On the board the direction is corrected live, so the notes should not pretend that the first version written there was final. What is stable is the cyclic structure. With the present convention,

$$Pe_1 = e_2, \qquad Pe_2 = e_3, \qquad Pe_3 = e_4, \qquad Pe_4 = e_1. \tag{1.19}$$

The same matrix acts on a general column by shifting its entries cyclically:

$$P \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} D \\ A \\ B \\ C \end{pmatrix}. \tag{1.20}$$

This is the lecture's first real worked example of an update rule written algebraically. A law of motion has become a matrix.

**Matrix multiplication as repeated updating.** Once a matrix represents one step of evolution, the reason for multiplying matrices by matrices becomes immediate. If

$$V' = MV, \tag{1.21}$$

then a second step gives

$$V'' = MV' = M^2 V. \tag{1.22}$$

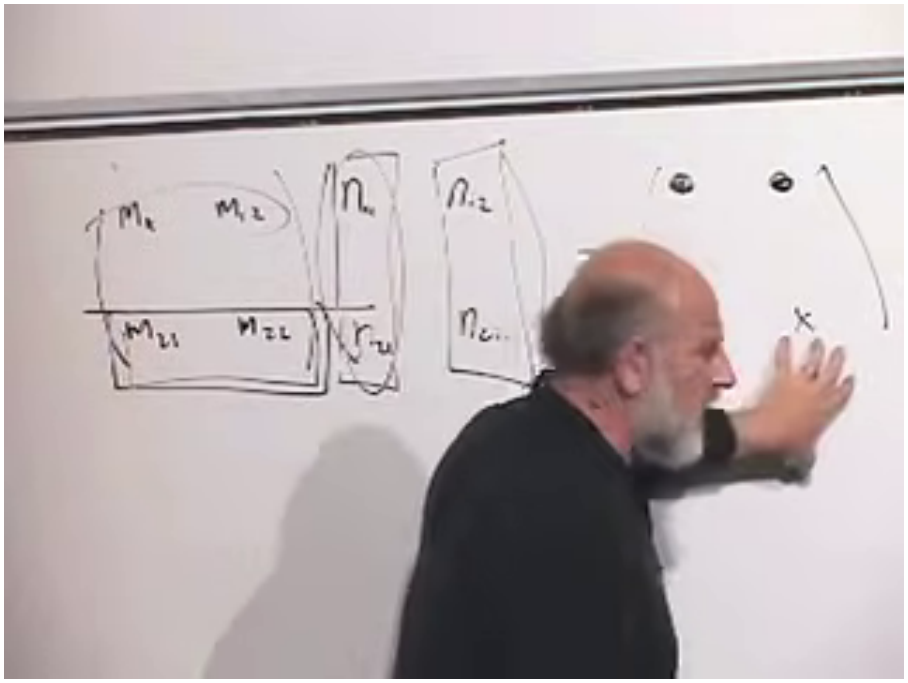


Figure 1.5: Board layout for the $2 \times 2$ matrix product. The pedagogical point is that the right-hand matrix may be read column by column, each column being acted on by the left-hand matrix.

So matrix multiplication composes updates. To evolve by two time steps, we apply the update matrix twice. To evolve by five, we multiply it into itself five times.

To make the arithmetic explicit, Susskind reduces the discussion to $2 \times 2$ matrices:

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}, \qquad N = \begin{pmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{pmatrix}. \tag{1.23}$$

Their product $C = MN$ is another matrix. The first entry is obtained by taking the first row of $M$ and multiplying it by the first column of $N$:

$$C_{11} = M_{11}N_{11} + M_{12}N_{21}. \tag{1.24}$$

The next entry comes from the first row of $M$ and the second column of $N$:

$$C_{12} = M_{11}N_{12} + M_{12}N_{22}. \tag{1.25}$$

The general rule is

$$C_{ij} = \sum_k M_{ik}N_{kj}. \tag{1.26}$$

The board layout in the lecture is especially revealing here. The right-hand matrix is drawn almost as two neighboring columns, because that is how Susskind wants us to think of it. Multiply the left matrix by the first column and put the result in the first column of the answer; multiply the left matrix by the second column and put the result in the second column of the answer. Matrix multiplication is thus visibly the same operation as matrix-times-vector multiplication, performed column by column.

**Row vectors multiplied by matrices.** The final pattern is the one obtained by multiplying a row vector by a matrix. If

$$B = (B_1, B_2, B_3, B_4), \tag{1.27}$$

then $BM$ is another row vector, and its $j$-th entry is found by multiplying the original row $B$ by the $j$-th column of $M$:

$$(BM)_j = \sum_i B_i M_{ij}. \tag{1.28}$$

This closes the elementary algebra of the lecture. A row times a column is a number. A matrix times a column is a column. A row times a matrix is a row. A matrix times a matrix is a matrix. Susskind ends by stressing that if we can do these manipulations fluently, then we already possess the basic mathematics that quantum mechanics will use over and over again.

Only at the end, in the questions, does the lecture phrase reversibility in matrix language. A reversible update should be represented by a matrix with an inverse. That is the algebraic shadow of the earlier physical statement that there must be a unique future and a unique past.

## 1.8 Summary

The first lecture does not yet present the new quantum rules. Instead it prepares the ground with unusual care. First we are told why abstraction is necessary: ordinary intuition is a classical inheritance, not a trustworthy guide to the microscopic world. Then the lecture reframes physics as information. From there it constructs classical bits, counts their states, and shows how surprisingly much of ordinary physics may be represented by long strings of binary data.

Only after that groundwork does the lecture turn from configuration to law. A physical system consists of possible states together with an update rule. Not every conceivable update rule is acceptable: fundamental laws must preserve the distinction between different pasts, which is why reversibility and uniqueness of past and future become central. The quantum name for the corresponding principle has already been previewed: unitarity.

Finally, after the break, the lecture installs the algebraic machinery required for the next step. Vectors, inner products, matrices, and matrix multiplication are not side remarks. They are the clean language of updating. So the lecture ends exactly where it should end: classical bits have been lifted into linear algebra, and the stage is set for qubits.

# Chapter 2

# Quantum Entanglements, Part 1

These notes follow Leonard Susskind's second lecture in the supplementary *Quantum Entanglement* track, in an edition curated by LazyingArt LLC. The lecture opens by leaving the world of classical bits and entering quantum mechanics at its simplest nontrivial point: a single electron spin. But it does not begin with formalism. It begins with a puzzle. The electron can be prepared as though it were pointing in many different directions, yet an up-down measurement returns only two answers. The chapter should be read in that order. We first let the classical picture build up enough force to fail, and only then do we introduce the complex vector space of states that can carry the physics.

## 2.1 Classical Bits, Qubits, and the Spin Example

We begin where the lecture begins: classical physics with a discrete state space. In that setting the space of states is a Boolean space, a collection of points, one point for each possible state, and the logic governing those states is ordinary classical logic. A classical bit is the simplest example. It has two possible values and nothing in between.

If we draw a two-state object with one end distinguished from the other, then "up" is one state and "down" is another. That is a classical bit in its barest form: two possibilities, no intermediate state. One can build many-bit systems out of such objects, and in classical physics that point of view goes a surprisingly long way.

Quantum mechanics forces a different notion of state, and Susskind chooses the simplest example immediately: the spin of an electron. He also warns us, at the outset, that the word *vector* is about to become dangerous.

On the one hand, there is the familiar vector in ordinary space: an arrow with magnitude and direction. To keep the distinction visible, we will call that a *spatial vector*. The electron carries such a spatial vector in the form of its magnetic moment. The magnetic moment has a fixed magnitude for every electron, but its direction may vary.

On the other hand, there is the abstract vector that belongs to a vector space of states. That vector is not an arrow in space. It is a mathematical object, and before long it will be this abstract vector, not the spatial arrow, that carries the state of the qubit.

That is already the lecture's first tension. The electron seems to come equipped with something that can point in any direction in space, but the measurement story we are heading toward will not

return a continuum of answers. We are being steered toward a conflict between classical imagery and quantum logic.

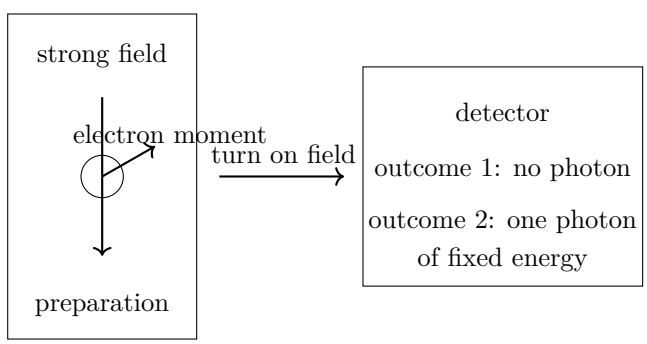
## 2.2 The Classical Preparation-and-Detection Story, and Why It Fails

Susskind next does something very deliberate: he tells a classical story in full. He even says that for the moment we are still thinking very classically. The point is not to endorse that story. The point is to let it fail in a way that makes the quantum alternative unavoidable.

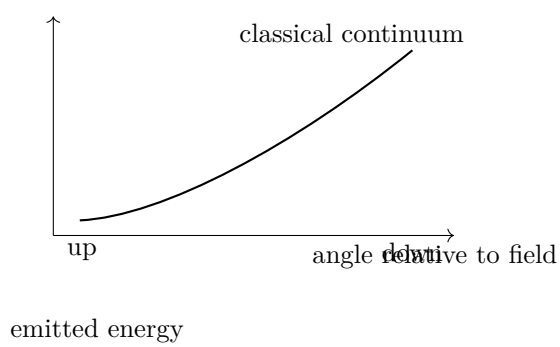
Imagine the electron as a tiny bar magnet with a north pole and a south pole. Suppose we place it in a strong external magnetic field. A classical magnet does not, in general, jump instantly into alignment. In vacuum, with no friction, it precesses around the external field. Then it radiates away its excess energy and settles into the energetically preferred orientation. That is the lecture's classical picture of *preparation*: choose the direction of the external field, wait long enough, and the electron's magnetic moment comes into equilibrium with that field.

The same setup then suggests a classical model of *detection*. Put the electron into a known field and watch how much radiation it emits as it relaxes. If the electron begins almost aligned with the field, only a little energy is emitted. If it begins farther away, more energy is emitted. If it begins anti-aligned, the emitted energy is maximal. So, classically, the emitted energy should be a continuous function of the initial angle.

The lecture's experimental staging is captured well enough by a schematic reconstruction:




In this classical picture we do *not* need an explicit formula for emitted energy as a function of angle, and the lecture never gives one. What matters is only the qualitative statement that the response is continuous.



This, however, is exactly the point at which the lecture reverses itself. Susskind says, in effect: that was the wrong story. For a real electron the detector does *not* produce a continuous output.

Once the magnetic field is turned on, only two things can happen:

- no photon is emitted;
- one photon of a definite energy is emitted, namely the energy corresponding to the transition from down to up.

Nothing in between ever appears. The detector does not report a small amount of up or a medium amount of down. It returns one of two answers.

What *does* depend on the preparation is the probability distribution over those two answers. If the electron has been prepared vertically upward and we measure it in the same up-down setup, then no photon is emitted. If it has been prepared vertically downward and we reverse the field appropriately, the photon is emitted with probability one. If it has been prepared horizontally and then measured in the vertical device, the two outcomes occur with equal probabilities:

$$P(\text{one photon}) = \frac{1}{2}, \qquad P(\text{no photon}) = \frac{1}{2}. \tag{2.1}$$

The lecture also pauses, in audience questions, over the operational meaning of this probability. A single run does not display a distribution. We must repeat the whole prepare-and-measure procedure many times. And once a given measurement has found the electron in the up state, repeating the same measurement does not recover the earlier preparation. The system has been fixed relative to that detector.

### 2.2.1 Question & Answer

**Question.** If the electron can be prepared in arbitrary directions, why does the up-down detector return only up or down?

**Answer.** Because the preparation direction does not create new outcomes for a fixed detector. A vertical detector always asks a binary question. The preparation changes only the probabilities of the two available answers. The mystery, then, is not that new detector values appear. The mystery is that a continuously variable preparation changes only a two-valued probability distribution.

That is the lecture's main puzzle. In one sense the electron seems to have only two measurable states. In another sense it can still be prepared in many directions. A set of two points will not be enough to describe such an object.

## 2.3 From the Puzzle to a Complex Vector Space of States

Now comes the lecture's central pivot. Susskind says, in substance, that we need a different mathematics of states. Not a Boolean set of two points, but a vector space.

He also tells us what the bargain is. We are going to do some abstract mathematics first and interpret it afterward. The first piece of that mathematics is the algebra of complex numbers.

We introduce the imaginary unit $i$ by

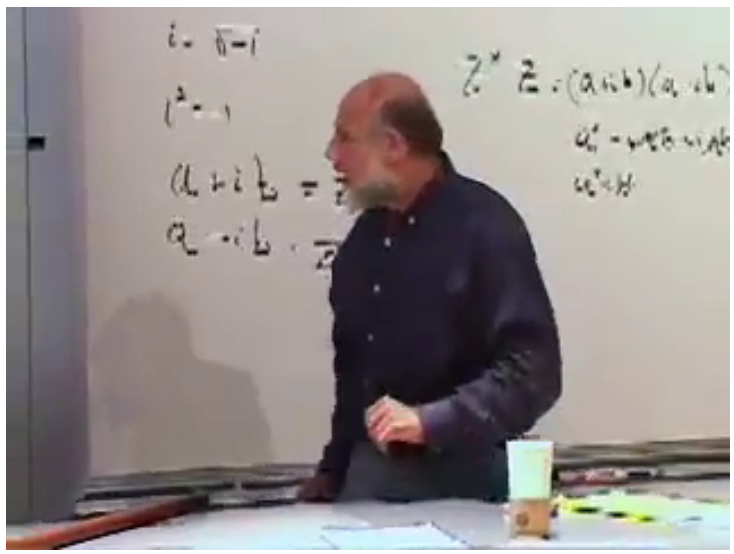
$$i^2 = -1. \tag{2.2}$$

A complex number is written

$$z = a + ib, \qquad z^* = a - ib, \tag{2.3}$$

where $a$ and $b$ are real and $z^*$ denotes the complex conjugate.

The lecture's blackboard cluster for this interlude is worth keeping, even though the frame is only partial evidence for the exact line-by-line algebra.



A key example is the product of a complex number with its conjugate:

$$\begin{aligned} z^* z &= (a + ib)(a - ib) && (2.4)\\ &= a^2 - iab + iab + b^2 && (2.5)\\ &= a^2 + b^2. && (2.6) \end{aligned}$$

The cross terms cancel, and the result is real and nonnegative. That is not yet quantum mechanics, but it is the algebraic pattern quantum mechanics will use when amplitudes become probabilities.

**Definition 2.1.** A complex vector space, in the minimal sense used in this lecture, is a collection of objects that can be added and multiplied by complex numbers, with the result remaining in the same collection.

The lecture then gives the simplest concrete realization of this idea: column vectors with complex entries. If

$$A \leftrightarrow \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix},$$

then multiplication by a complex number $c$ is defined componentwise:

$$c \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} cA_1 \\ cA_2 \\ cA_3 \end{pmatrix}, \tag{2.7}$$

and addition is defined by

$$\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} + \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix} = \begin{pmatrix} A_1 + B_1 \\ A_2 + B_2 \\ A_3 + B_3 \end{pmatrix}. \tag{2.8}$$

There is no need yet to say more. The point is to establish the kind of space in which quantum states will live. For the spin of a single electron, the relevant case will be a two-dimensional complex vector space.

## 2.4 Column Vectors, Bras, Kets, and the Inner Product

The lecture now tightens the notation. Abstract vectors are represented by columns, and their conjugates by rows.

If

$$|a\rangle \leftrightarrow \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$

then its conjugate is represented by

$$\langle a| \leftrightarrow \begin{pmatrix} a_1^* & a_2^* \end{pmatrix}.$$

The star is implicit in turning the ket into the bra; that is exactly the point Susskind pauses over in the lecture.

The notation $\langle b|a\rangle$ is a *bracket* split into its two halves, bra and ket. In components the rule is

$$\langle b|a\rangle = b_1^* a_1 + b_2^* a_2. \tag{2.9}$$

This is the inner product.

If we take the inner product of a vector with itself, we find

$$\langle a|a\rangle = a_1^* a_1 + a_2^* a_2. \tag{2.10}$$

Each term has the form $z^* z$, and therefore each term is real and nonnegative. So $\langle a|a\rangle$ itself is real and nonnegative.

*Remark* 2.2. The lecture interprets $\langle a|a\rangle$ as the square of the size of the vector. We are not yet using this abstractly; we are preparing it for a physical interpretation.

This is the point at which the mathematics begins to justify itself. Complex numbers were introduced because their conjugation law produces positive real quantities. Those are exactly the quantities that can later be interpreted as probabilities.

## 2.5 Up and Down as an Orthogonal Basis of States

At this point Susskind explicitly returns from the abstract vector space to the quantum bit. The detector has two possible outputs. Let us denote the corresponding states by

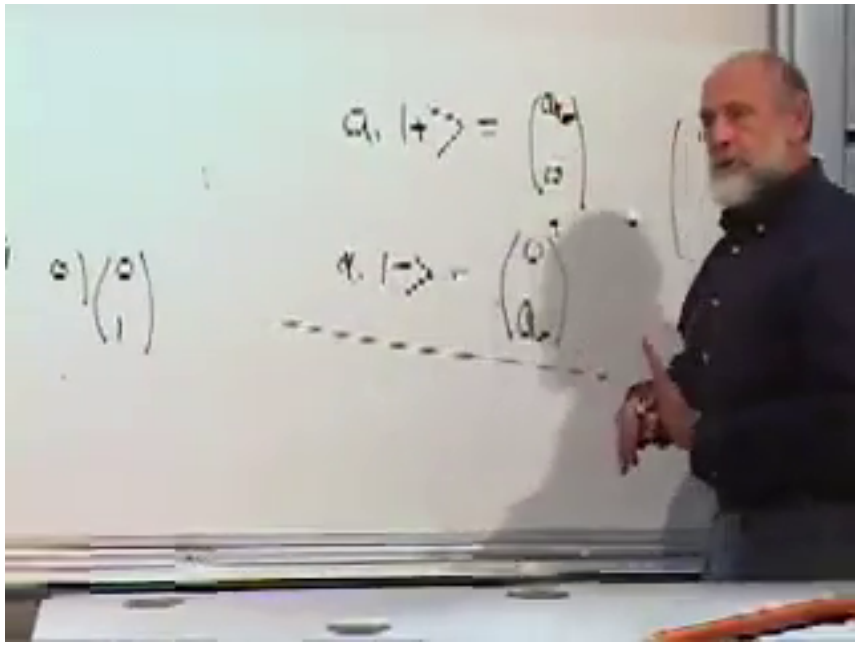
$$|+\rangle \qquad \text{and} \qquad |-\rangle.$$

These stand for the electron being found up or down in the chosen measurement basis.

Now the lecture makes the space concrete. A general state of the electron spin is written as a linear combination

$$|\psi\rangle = a_+|+\rangle + a_-|-\rangle. \tag{2.11}$$

The corresponding blackboard layout is an important piece of visual evidence because it shows exactly how the coefficient-weighted basis states are assembled into a single column vector.



Once the basis has been fixed, the canonical column-vector representatives are

$$|+\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.12}$$

Then

$$a_+|+\rangle \leftrightarrow \begin{pmatrix} a_+ \\ 0 \end{pmatrix}, \qquad a_-|-\rangle \leftrightarrow \begin{pmatrix} 0 \\ a_- \end{pmatrix},$$

and adding these two contributions gives

$$|\psi\rangle \leftrightarrow \begin{pmatrix} a_+ \\ a_- \end{pmatrix}. \tag{2.13}$$

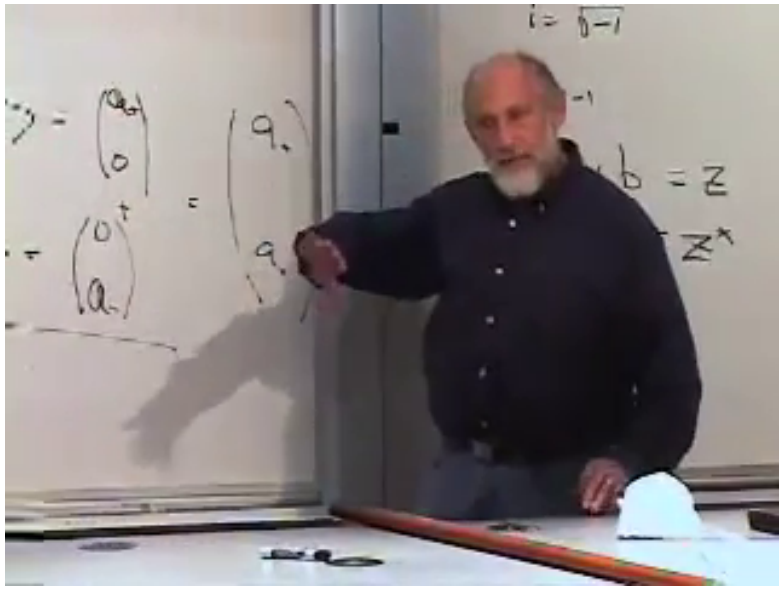
The lecture later revisits the same identification more cleanly on the board, and that cleaner basis-fixing picture should remain visible as well.

Now comes the physical interpretation. The squared magnitudes of the coefficients are the probabilities for the two detector outcomes:

$$P(+) = a_+^* a_+ = |a_+|^2, \qquad P(-) = a_-^* a_- = |a_-|^2. \tag{2.14}$$

Since these are the only two possibilities, they must add to one:

$$|a_+|^2 + |a_-|^2 = 1. \tag{2.15}$$



Equivalently,

$$\langle\psi|\psi\rangle = 1. \tag{2.16}$$

This is the normalization condition. The physical states are the normalized vectors in the vector space.

The two basis states are orthonormal:

$$\langle +|+\rangle = 1, \qquad \langle -|-\rangle = 1, \tag{2.17}$$

$$\langle +|-\rangle = 0, \qquad \langle -|+\rangle = 0. \tag{2.18}$$

The lecture explicitly checks these relations in components. For example,

$$\langle +|-\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

So the up and down states correspond to orthogonal vectors in the state space.

**Worked example.** The lecture does not move too quickly past its first nontrivial example. It writes, in effect,

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

then immediately notices that this cannot yet be a physical state because its norm-squared is 2. To repair it, we divide by $\sqrt{2}$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{2.19}$$

Now the probabilities are

$$P(+) = \frac{1}{2}, \qquad P(-) = \frac{1}{2}.$$

This is the lecture's equal-weight state. It is not pure up and not pure down. It is one of the states associated with a horizontal preparation.

We now see how the lecture wants us to think. A detector aligned along a chosen axis still returns only two answers. But the state space is larger than those two detector values, because it contains all normalized linear combinations of the corresponding basis vectors.

## 2.6 What the State Vector Does and Does Not Encode

After the break Susskind performs a reset. More than one person is probably confused, he says, about the two notions of vector. That is exactly the right place to slow down and separate them.

A spatial vector in ordinary space has three real components. A two-component state vector,

$$\begin{pmatrix} a_+ \\ a_- \end{pmatrix},$$

has two complex components, which means four real numbers before any constraints are imposed. So the state vector is not just a disguised spatial arrow.

Of course, not all four of those real numbers are physically independent. Normalization removes one. Overall phase removes another. But the lecture does not yet build the full geometry that connects state vectors to spatial directions, and we should not force it. The right statement, and the lecture's own statement, is simpler: the two objects are related, but not simply related.

There is another important clarification here. The dimension of the state space is fixed experimentally. The up-down measurement has two possible outcomes, and that is why we use a two-dimensional complex vector space. In that sense the mathematics is not arbitrary. The experiment tells us how many basis states we need.

### 2.6.1 Question & Answer

**Question.** Is $-|+\rangle$ the same thing as $|-\rangle$?

**Answer.** No. The symbol $-$ in $|-\rangle$ is part of a label. It does not mean minus one times the plus state. By contrast, $-|+\rangle$ is an ordinary scalar multiple of $|+\rangle$, and it is perfectly allowed in the vector space.

This answer leads immediately to the lecture's next point. The states $|+\rangle$ and $-|+\rangle$ are not equal as vectors, but they represent the same physics, because an overall phase does not change the probabilities of any measurement. More generally, if $|\psi\rangle$ is multiplied by a complex number of magnitude one, the physical state is unchanged.

But a relative sign inside a superposition is a different matter. The states

$$a_+|+\rangle + a_-|-\rangle \qquad \text{and} \qquad a_+|+\rangle - a_-|-\rangle$$

need not represent the same physics. The lecture does not yet unpack the full meaning of relative phase, but it clearly insists on this distinction: overall phase is inessential, relative phase can matter.

## 2.7 From Classical Observables to Quantum Matrices

The lecture now shifts its center of gravity. Up to now we have been asking what a state is. Next we ask what a measurable quantity is.

In classical mechanics, if the possible states are discrete points labeled by $n$, then an observable is simply a function that assigns a real number to each state:

$$f_n.$$

If the system is not known exactly, but only through a probability distribution $p_n$, then the average value of the observable over many repeated experiments is

$$\langle f \rangle = \sum_n p_n f_n. \tag{2.20}$$

The lecture lingers here because it wants matrices to emerge by motivation rather than fiat. A coin provides the simplest example. Suppose heads is assigned the value $+1$ and tails the value $-1$. If the probabilities are $p_+$ and $p_-$, then

$$\langle f \rangle = p_+ - p_-. \tag{2.21}$$

This is the average value of the observable, not necessarily an attainable single-shot outcome.

### 2.7.1 Question & Answer

**Question.** How can the average value be zero when zero is not one of the actual outcomes?

**Answer.** Because the average value is not the same thing as a detector reading from a single run. If a fair coin yields $+1$ half the time and $-1$ half the time, then the average over many trials is zero, even though no single toss ever produces zero. The average is a summary of the distribution, not a new observable value that occurs in one trial.

The lecture then singles out a special classical observable: the indicator function that is 1 on one state and 0 on all the others. In that case the average is exactly the probability of that state. So if we know how to compute average values of all observables, we also know how to recover probabilities.

This now raises the quantum question: what mathematical objects represent observables in quantum mechanics? The answer, for the present finite-dimensional setting, is matrices.

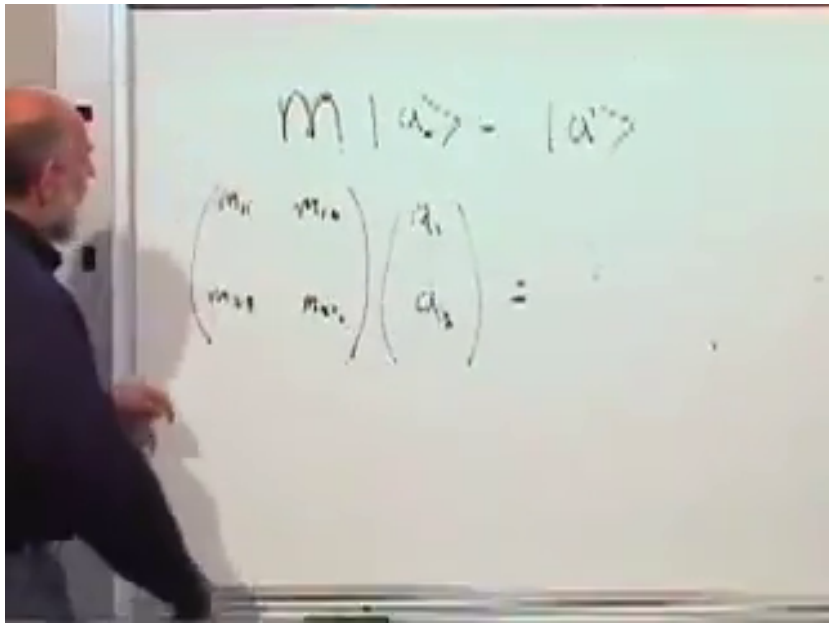
A matrix acts on a state vector and produces another vector:

$$M|a\rangle = |a'\rangle. \tag{2.22}$$

The lecture's board picture should remain in place here because it preserves the two-level transition from abstract operator language to explicit component calculation.

If

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}, \qquad |a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$



then the component rule is

$$\begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} m_{11}a_1 + m_{12}a_2 \\ m_{21}a_1 + m_{22}a_2 \end{pmatrix}. \tag{2.23}$$

Susskind first motivates this by going back to a very ordinary real vector space, the plane of arrows drawn on the board. In that setting matrices represent linear transformations:

- $$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$
  doubles every vector;
- $$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$
  stretches only the $y$-component;
- $$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
  reflects across the diagonal $x = y$;
- $$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$
  rotates the plane by $90°$, in the clockwise convention used in the discussion.

The lesson is simple: matrices are the language of linear transformations of vector spaces. That is why they are the right candidates for quantum observables.

Now comes the special class singled out by the lecture.

**Definition 2.3.** A matrix $M$ is Hermitian if

$$M_{ij} = M^*_{ji}. \tag{2.24}$$

This is a kind of reality condition, though it does *not* say that every entry is real. It says, first, that on the diagonal

$$M_{ii} = M^*_{ii},$$

so diagonal entries are real. It says, second, that off-diagonal entries come in complex-conjugate pairs. In the special case that all entries are real, Hermitian reduces to symmetric.

The lecture ends this part of the story with a clear declaration. In classical mechanics observables are functions on state points. In quantum mechanics observables will be represented by Hermitian operators, which at the moment means Hermitian matrices.

There is one further preview. If we want to *kick* the electron, that is not generally done by a Hermitian operator. It is done by a unitary operator. At this stage only one fact is stated: unitary operators preserve the length of vectors. Their role in dynamics and physical operations belongs to what comes next.

## 2.8 Summary

The lecture has carried us from a concrete spin-measurement puzzle to the threshold of the quantum formalism. We began with a classical bit and a classical magnetic-arrow picture of the electron. We let that picture generate a continuous detector response, and then watched the lecture reject it. A real electron, measured in a fixed up-down setup, yields only two outcomes. What varies with preparation is not the list of outcomes but the probability distribution over them.

That failure of the classical story forced a new mathematics of states. We introduced complex numbers, conjugation, column vectors, bras, kets, and the inner product. We chose the orthonormal basis $|+\rangle$ and $|-\rangle$, wrote a general qubit state as

$$|\psi\rangle = a_+|+\rangle + a_-|-\rangle,$$

and interpreted $|a_+|^2$ and $|a_-|^2$ as probabilities, subject to normalization.

After that, the lecture shifted from states to observables. Classical observables are functions $f_n$ on discrete states, with averages

$$\langle f\rangle = \sum_n p_n f_n.$$

Quantum observables are more intricate: they are represented by matrices acting on state vectors, with Hermitian matrices singled out as the measurable ones. That is the mathematical entrance fee the lecture insists upon before entanglement can be discussed honestly. Once states, inner products, and Hermitian operators are in hand, the rest of the structure of quantum mechanics can be built without sleight of hand.

# Chapter 3

# Lecture 3: Quantum Entanglements, Part 1

We do not begin with entanglement itself. We begin where the lecture begins: with the small mathematical facts without which the subject becomes empty language. Susskind wants everyone on the same page, and so we first clear the runway with complex numbers, normalized state vectors, Hermitian observables, eigenvalues and eigenvectors, and the probability rule for spin. The guiding idea throughout is that quantum mechanics is a calculus for probabilities. These companion notes follow Leonard Susskind's lecture closely, with curation by LazyingArt LLC, and keep the two-level spin system in view at every stage.

## 3.1 Complex Numbers, Phases, and Conjugation

The opening review is brisk on purpose. We are not doing complex analysis for its own sake; we are fixing the notation that quantum mechanics will immediately require.

A complex number may be written in Cartesian form,

$$z = x + iy, \tag{3.1}$$

or in polar form,

$$x = r\cos\theta, \qquad y = r\sin\theta, \tag{3.2}$$

so that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}. \tag{3.3}$$

The point of the notation $e^{i\theta}$ is that it behaves like an exponential:

$$e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}. \tag{3.4}$$

This is nothing but the angle-addition formulas in compressed form.

Complex conjugation reverses the sign of the imaginary part:

$$z^* = x - iy. \tag{3.5}$$

Applied to a pure phase,

$$(e^{i\theta})^* = e^{-i\theta}, \tag{3.6}$$

and therefore

$$e^{i\theta}e^{-i\theta} = 1. \tag{3.7}$$

More generally,

$$zz^* = x^2 + y^2. \tag{3.8}$$

So the numbers of the form $e^{i\theta}$ lie on the unit circle:

$$|e^{i\theta}|^2 = 1. \tag{3.9}$$

The lecture uses the informal phrase "unitary numbers" in passing, but the mathematically stable term is *pure phase*. Every complex number is a positive real modulus times a pure phase. That is all we need from this opening review, but we will need it at once.

## 3.2 States as Normalized Complex Vectors

Now comes the hard pivot. Quantum mechanics is introduced as a calculus for calculating probabilities. Probabilities of what? Probabilities for measurement outcomes. The things that are measured are observables, and the systems being measured are prepared in states.

In the simple two-level system, a general state is written as

$$|\psi\rangle = \alpha|u\rangle + \beta|d\rangle, \tag{3.10}$$

where $\alpha$ and $\beta$ are complex numbers. In the standard basis,

$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.11}$$

If we measure up or down, the probabilities are

$$P(u) = |\alpha|^2, \qquad P(d) = |\beta|^2. \tag{3.12}$$

Because the total probability must be one, the state is normalized:

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1. \tag{3.13}$$

This is the first postulate in the lecture's presentation: states are vectors in a complex vector space, and we normalize them. In a higher-dimensional example the same pattern persists. The basis vectors are columns with a single nonzero entry, and an arbitrary state is a complex linear combination of them.

At this point Susskind pauses on an important issue. The probabilities $|\alpha|^2$ and $|\beta|^2$ do not completely determine the complex amplitudes. Multiplying by a pure phase changes the number but not its magnitude:

$$|e^{i\theta}\alpha|^2 = |\alpha|^2. \tag{3.14}$$

So the amplitudes contain more information than probabilities alone. The lecture does not yet say what that extra information is for; it merely marks that there is something still to be explained.

There is another warning already in place. These state vectors are not ordinary arrows in real space. The vectors $|u\rangle$ and $|d\rangle$ are orthogonal in the space of states. Later we will also speak about spatial directions of spin and magnetic moment, and those are different objects. The distinction must be kept in mind from the beginning.

## 3.3 Observables, Transpose, and Hermitian Matrices

What about the quantities that we measure? The lecture now turns to observables. The outcomes of measurements are always real numbers, or at any rate can always be reduced to real numbers. That requirement is what narrows the matrix language.

Before defining the relevant class of matrices, Susskind introduces transpose and then Hermitian conjugation.

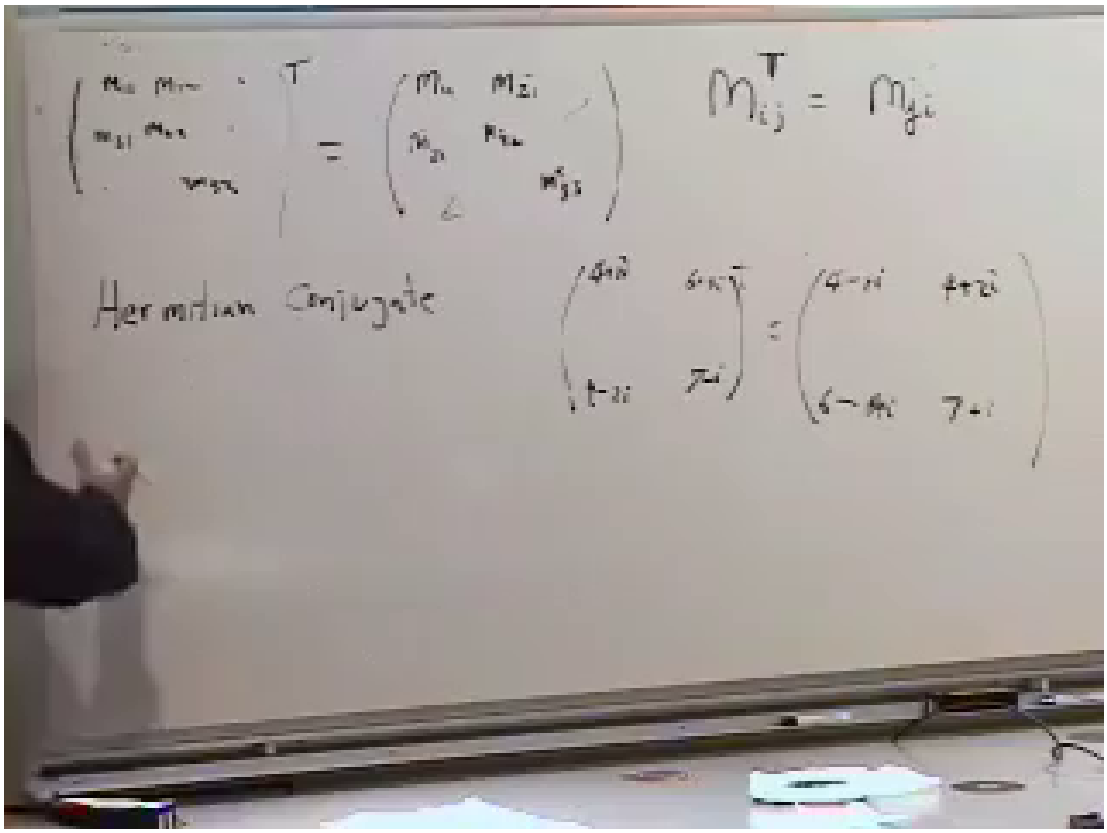



Figure 3.1: Matrix transpose and symmetry rule.

The board first displays transpose concretely. For a $3 \times 3$ matrix,

$$\begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}^T = \begin{pmatrix} m_{11} & m_{21} & m_{31} \\ m_{12} & m_{22} & m_{32} \\ m_{13} & m_{23} & m_{33} \end{pmatrix}. \tag{3.15}$$

In index notation,

$$(M^T)_{ij} = M_{ji}. \tag{3.16}$$

The effect is simply to interchange rows and columns, or if one prefers a geometric picture, to reflect across the main diagonal.

A symmetric matrix is one that is unchanged by this operation:

$$M_{ij} = M_{ji}. \tag{3.17}$$

That is the content of the board equation near the figure, and it is worth preserving because the lecture uses this simpler notion as a warm-up before turning to the complex case.

Now we add complex conjugation. The Hermitian conjugate is defined by

$$M^\dagger = (M^T)^*. \tag{3.18}$$

A Hermitian matrix is one equal to its own Hermitian conjugate:

$$M = M^\dagger. \tag{3.19}$$

In components this becomes

$$M_{ij} = (M_{ji})^*. \tag{3.20}$$

The transcript is garbled right at this point, but this is the clean condition the lecture is reaching for.

The consequences are immediate. The diagonal entries are real, because on the diagonal we have $i = j$, so the entry must equal its own complex conjugate. Off the diagonal, reflected entries are complex conjugates of one another. The right-hand board example is not sharp enough for literal transcription, so it is better to write the standard $2 \times 2$ form explicitly:

$$M = \begin{pmatrix} a & z \\ z^* & b \end{pmatrix}, \qquad a, b \text{ real}. \tag{3.21}$$

This is the matrix version of the reality condition appropriate to observables.

### 3.3.1 Question & Answer

**Question.** If observables are real, why are we using complex numbers at all?

**Answer.** The lecture deliberately refuses to answer this too early. Anything we literally measure can be encoded by real numbers. One may, if one wishes, package several real quantities into a complex number, but that is not the real issue. The real question is: why is quantum mechanics *driven* to complex amplitudes? Susskind postpones the answer. It belongs to time evolution and reversibility, not to the static postulates being organized here. For the moment we keep the complex vector space in place and move on.

## 3.4 Matrix Elements and Real Expectation Values

The lecture now writes down a few very elementary identities and immediately extracts the first genuine payoff. These are the "theorems without names" that justify the Hermitian definition.

First, the inner product satisfies

$$\langle B|A\rangle = \langle A|B\rangle^*. \tag{3.22}$$

Second, for any matrix $M$,

$$\langle B|M|A\rangle = \big(\langle A|M^\dagger|B\rangle\big)^*. \tag{3.23}$$

If $M$ is Hermitian, so that $M^\dagger = M$, this becomes

$$\langle B|M|A\rangle = \big(\langle A|M|B\rangle\big)^*. \tag{3.24}$$

The quantity $\langle B|M|A\rangle$ is the matrix element of $M$ between the states $|A\rangle$ and $|B\rangle$.

**Proposition 3.1.** *If $M$ is Hermitian, then $\langle A|M|A\rangle$ is real.*

*Proof.* Set $B = A$ in the Hermitian matrix-element identity. Then

$$\langle A|M|A\rangle = \big(\langle A|M|A\rangle\big)^*. \tag{3.25}$$

A number equal to its own complex conjugate is real. Therefore $\langle A|M|A\rangle \in \mathbb{R}$. □

The lecture then gives this object its standard name: the *expectation value* of $M$ in the state $|A\rangle$. The probabilistic interpretation of the name is deferred, but the mathematical point is already decisive. If an observable is to have real measurement outcomes, its average value must certainly be real. Hermitian operators have exactly that property in every state. That is the first serious reason they are the right representatives for observables.

## 3.5 Eigenvalues and Eigenvectors from the Simplest Examples

At this point the lecture insists that we continue doing mathematics. One can speak vaguely about entanglement and leave with a mystical feeling, but not with understanding. So the next concept, and essentially the last big mathematical concept needed for the day, is the notion of eigenvalues and eigenvectors.

**Definition 3.2.** A nonzero vector $|a\rangle$ is an eigenvector of $M$ with eigenvalue $\lambda_a$ if

$$M|a\rangle = \lambda_a|a\rangle. \tag{3.26}$$

For a Hermitian matrix the eigenvalue must be real.

**Proposition 3.3.** *If $M$ is Hermitian and $M|a\rangle = \lambda_a|a\rangle$, then $\lambda_a$ is real.*

*Proof.* Multiply the eigenvalue equation on the left by $\langle a|$:

$$\langle a|M|a\rangle = \lambda_a\langle a|a\rangle. \tag{3.27}$$

The left-hand side is real because $M$ is Hermitian, and $\langle a|a\rangle$ is real and positive for a nonzero vector. Therefore

$$\lambda_a = \frac{\langle a|M|a\rangle}{\langle a|a\rangle} \tag{3.28}$$

is real. □

Now comes the lecture's first great interpretive statement. If $M$ represents an observable, then the values one can actually measure are the eigenvalues of $M$. The eigenvector associated with a given eigenvalue is the state in which that value is obtained with certainty.

To make this statement feel unavoidable, Susskind begins with the easiest possible examples.

Take the diagonal matrix

$$M = \begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix}. \tag{3.29}$$

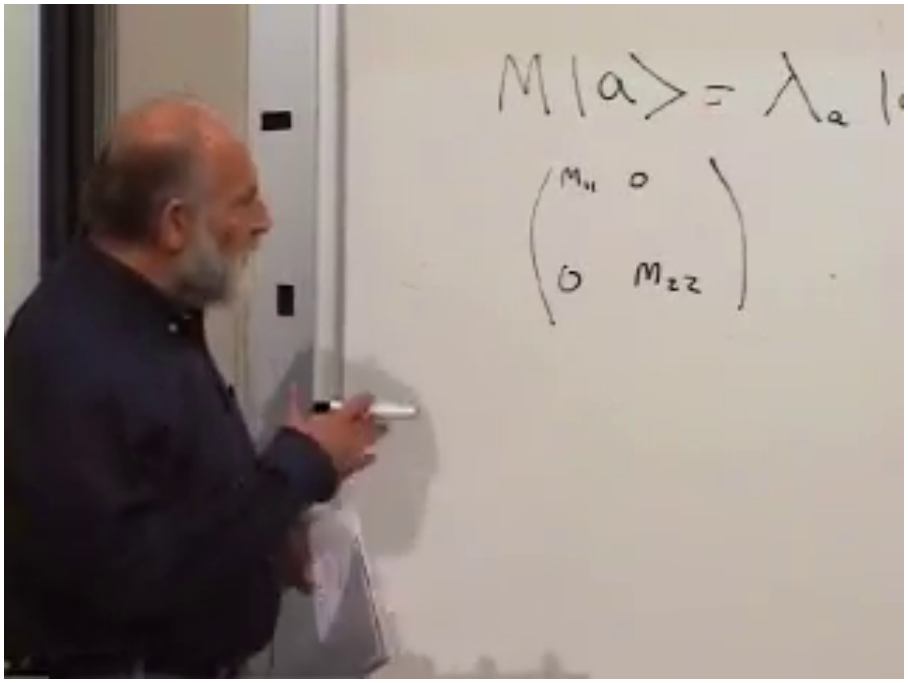



Figure 3.2: Eigenvector equation with a diagonal matrix.

The first basis vector is immediately an eigenvector:

$$\begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = m_{11} \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{3.30}$$

So $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ has eigenvalue $m_{11}$.

Likewise,

$$\begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = m_{22} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.31}$$

So $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ has eigenvalue $m_{22}$.

These are almost trivial calculations, but the lecture uses them to establish the rhythm we will need later. Matrix times vector equals number times the same vector. The number is a possible measurement value. The vector is a state of certainty for that measurement. One can already see the pattern for larger diagonal matrices: the diagonal entries are the eigenvalues, and the standard basis vectors are the eigenvectors.

## 3.6 Pauli Matrices and the Two-Level Spin System

Only after the diagonal examples are under control does the lecture turn to the actual two-state spin observables.

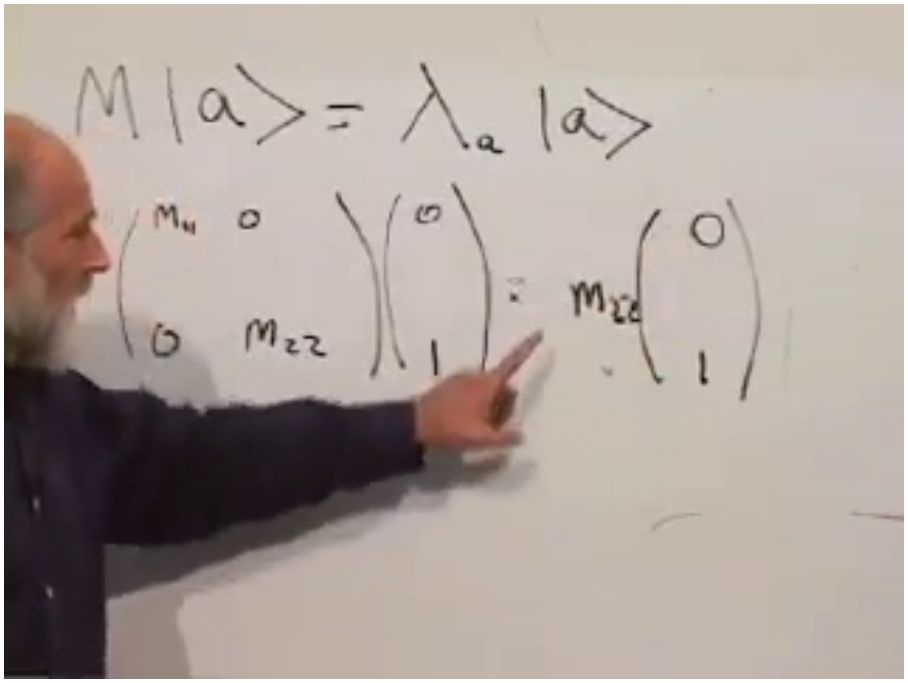



Figure 3.3: Diagonal matrix acting on the lower basis vector.

The first one is

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.32}$$

Its eigenvectors are the familiar up and down states:

$$\sigma_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \sigma_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = - \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.33}$$

So the possible measured values are $+1$ and $-1$, and the corresponding states are precisely the states of definite spin along the 3-axis.

The second matrix is the off-diagonal example:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{3.34}$$

Here the lecture first writes the candidate eigenvector and only then asks what the matrix does to it. The screenshot captures that pedagogical moment.

The candidate vector

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{3.35}$$

is indeed an eigenvector, but not normalized. In quantum mechanics we normalize the state, so we use

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{3.36}$$

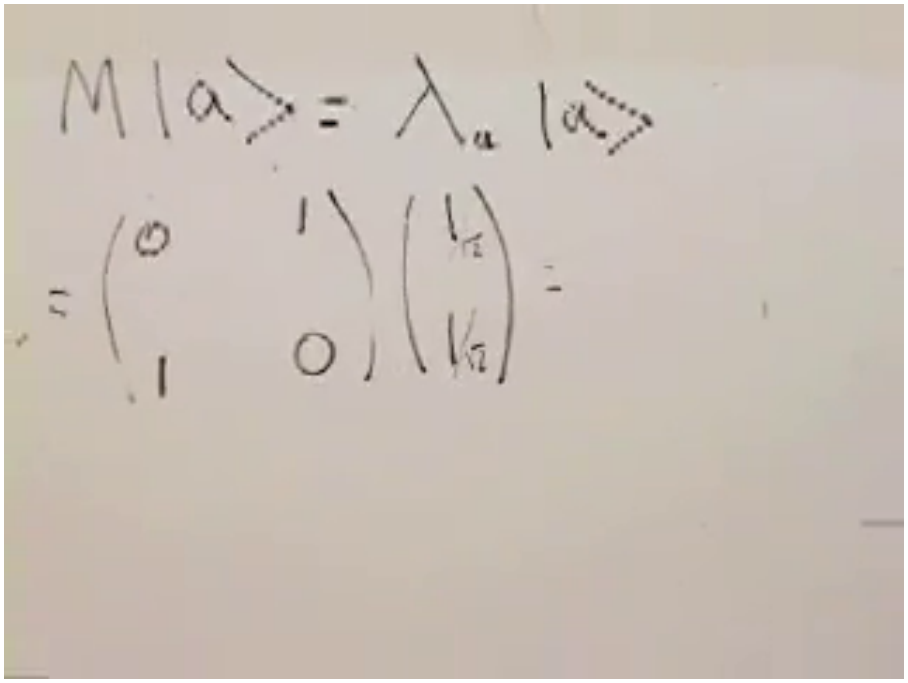



Figure 3.4: Off-diagonal matrix acting on an equal-weight vector.

The right-hand side is not yet written on the board in the frame, so we supply it from the surrounding calculation:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{3.37}$$

So this is an eigenvector of $\sigma_1$ with eigenvalue $+1$. The companion vector

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{3.38}$$

satisfies

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{3.39}$$

The lecture pauses here to emphasize something subtle but basic. If a vector is an eigenvector, multiplying it by any nonzero scalar leaves it an eigenvector. Normalization is not part of the eigenvector condition itself. But once the vector is to represent a physical state, we normalize it.

There is also a brief geometric aside in the lecture. One can view $\sigma_1$ as interchanging the two basis directions $E_1$ and $E_2$. In that picture the diagonal directions are singled out, and that is why the eigenvectors of $\sigma_1$ point along $(1, 1)$ and $(1, -1)$.

The lecture then introduces the third Pauli matrix,

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{3.40}$$

but postpones its eigenvectors for a little while. What matters immediately is that all three Pauli matrices share the same basic spectral pattern. They all square to the identity:

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = 1, \tag{3.41}$$

where 1 denotes the identity matrix. For example,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3.42}$$

If the square of an observable is the identity, then the square of any eigenvalue must be one. Therefore the only possible measured values are

$$\lambda = \pm 1. \tag{3.43}$$

This is the recurring fact the lecture wants us to absorb: different spin observables on the same two-dimensional state space have different eigenvectors, but the same two possible outcomes.

## 3.7 Orthogonality, Probabilities, and Spin Along an Arbitrary Axis

Now Susskind announces a fundamental theorem. Before proving it, he says what it means. If an observable can distinguish two states by assigning them different definite outcomes, then those states must be orthogonal. In the lecture's language: orthogonal states are measurably distinct states.

**Theorem 3.4.** *Let $M$ be Hermitian, and suppose*

$$M|A\rangle = \lambda_A|A\rangle, \qquad M|B\rangle = \lambda_B|B\rangle, \tag{3.44}$$

*with $\lambda_A \neq \lambda_B$. Then*

$$\langle B|A\rangle = 0. \tag{3.45}$$

*Proof.* Multiply the first equation on the left by $\langle B|$:

$$\langle B|M|A\rangle = \lambda_A\langle B|A\rangle. \tag{3.46}$$

Multiply the second equation on the left by $\langle A|$:

$$\langle A|M|B\rangle = \lambda_B\langle A|B\rangle. \tag{3.47}$$

Now complex-conjugate the second equation. Since $M$ is Hermitian and $\lambda_B$ is real,

$$\langle B|M|A\rangle = \lambda_B\langle B|A\rangle. \tag{3.48}$$

Comparing the two equations gives

$$(\lambda_A - \lambda_B)\langle B|A\rangle = 0. \tag{3.49}$$

If $\lambda_A \neq \lambda_B$, the only possibility is $\langle B|A\rangle = 0$. □

The lecture then checks this theorem on the $\sigma_1$ example. The two normalized eigenvectors are

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \tag{3.50}$$

and their inner product is

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0. \tag{3.51}$$

So the theorem is not merely formal; it is already visible in the simplest example.

At this point the lecture supplies the promised eigenvectors of $\sigma_2$:

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}. \tag{3.52}$$

One has eigenvalue $+1$, the other eigenvalue $-1$. So $\sigma_1$, $\sigma_2$, and $\sigma_3$ are all similar in the sense emphasized by the lecture: they all have the same two eigenvalues, $+1$ and $-1$, even though their eigenvectors differ.

Now comes the probability postulate. Suppose the system is prepared in a normalized state $|B\rangle$, and suppose $|A\rangle$ is a normalized eigenvector of an observable $M$ with eigenvalue $\lambda_A$. Then

$$P(\lambda_A|B) = |\langle A|B\rangle|^2 = \langle A|B\rangle\langle B|A\rangle. \tag{3.53}$$

This is the bridge between the spectral information and actual experimental prediction.

The lecture works examples immediately.

First prepare the electron in the $+1$ eigenstate of $\sigma_3$:

$$|B\rangle = \begin{pmatrix}1\\0\end{pmatrix}. \tag{3.54}$$

Now measure $\sigma_1$. The $+1$ eigenstate of $\sigma_1$ is

$$|A\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}. \tag{3.55}$$

Then

$$\langle A|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}, \tag{3.56}$$

$$P(\sigma_1 = +1 \,|\, \sigma_3 = +1) = |\langle A|B\rangle|^2 = \frac{1}{2}. \tag{3.57}$$

If we choose instead the $-1$ eigenvector of $\sigma_1$,

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \tag{3.58}$$

we again get probability $1/2$. So a state polarized along the 3-axis gives equal chances for the two outcomes when we measure along the 1-axis.

The lecture then reverses the order of preparation and measurement, and gets the same answer. That is not an accident of wording; it reflects the symmetry of $|\langle A|B\rangle|^2$.

Finally, the lecture takes a genuinely complex example. Prepare the system in the $+1$ eigenstate of $\sigma_1$:

$$|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \tag{3.59}$$

and measure $\sigma_2$. For the $+1$ eigenstate of $\sigma_2$, take

$$|A\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}. \tag{3.60}$$

Then

$$\langle A|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix} = \frac{1}{2} - \frac{i}{2}, \tag{3.61}$$

$$P(\sigma_2 = +1 \,|\, \sigma_1 = +1) = \left|\frac{1}{2} - \frac{i}{2}\right|^2 = \frac{1}{2}. \tag{3.62}$$

Again the result is one-half. The lecture uses these examples to train the eye: changing the measurement axis changes the probabilities, not the set of possible values.

### 3.7.1 Question & Answer

**Question.** Is this just the cosine squared of an angle? And are perpendicular pointers the same thing as orthogonal states?

**Answer.** Only if one keeps the analogy under tight control. For ordinary real unit vectors in three-dimensional space, the dot product is the cosine of the angle between them. But here the inner product of two states is generally complex, and the probability is its absolute square. So the ordinary geometric picture is at best an analogy.

The more important point is the distinction between state vectors and spatial directions. A spin direction in ordinary space is a *pointer*. A state vector lives in Hilbert space. The state orthogonal to spin-up along a given axis is spin-down along that same axis; it is not spin-up along a perpendicular spatial direction. That is why the lecture repairs the vocabulary explicitly: *perpendicular* belongs to ordinary-space pointers, while *orthogonal* belongs to states.

With that clarification made, the lecture turns to the component of spin along an arbitrary spatial direction. Here we leave Hilbert-space vectors for a moment and speak classically. Let

$$\hat{n} = (n_1, n_2, n_3), \qquad n_1^2 + n_2^2 + n_3^2 = 1. \tag{3.63}$$

This is a unit pointer in ordinary three-dimensional space. Classically the component of a spin pointer $\sigma$ along $\hat{n}$ is the dot product $\sigma \cdot \hat{n}$. The quantum candidate is therefore

$$\sigma \cdot \hat{n} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3. \tag{3.64}$$

Using the Pauli matrices,

$$\sigma \cdot \hat{n} = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}. \tag{3.65}$$

This is Hermitian at once: the diagonal entries are real, and the off-diagonal entries are complex conjugates.

The lecture then asks whether the square of this operator is again the identity. The key point is that when we multiply matrices, order matters. So the cross terms must be written in both orders. The Pauli matrices satisfy the anticommutation relations

$$\sigma_1\sigma_2 + \sigma_2\sigma_1 = 0, \qquad \sigma_2\sigma_3 + \sigma_3\sigma_2 = 0, \qquad \sigma_1\sigma_3 + \sigma_3\sigma_1 = 0. \tag{3.66}$$

Together with $\sigma_i^2 = 1$, this gives

$$\begin{aligned}(\sigma \cdot \hat{n})^2 &= (n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3)^2 \qquad (3.67)\\ &= n_1^2\sigma_1^2 + n_2^2\sigma_2^2 + n_3^2\sigma_3^2 \\ &\quad + n_1n_2(\sigma_1\sigma_2 + \sigma_2\sigma_1) + n_2n_3(\sigma_2\sigma_3 + \sigma_3\sigma_2) + n_1n_3(\sigma_1\sigma_3 + \sigma_3\sigma_1) \\ &= (n_1^2 + n_2^2 + n_3^2)\,1 \qquad (3.68)\\ &= 1.\end{aligned}$$

Therefore any eigenvalue $\lambda$ of $\sigma \cdot \hat{n}$ satisfies

$$\lambda^2 = 1, \tag{3.69}$$

and so

$$\lambda = \pm 1. \tag{3.70}$$

This is how the lecture closes. Along every axis, the possible measured values are still only $+1$ and $-1$. The quantum strangeness lies not in a changing spectrum, but in the changing probabilities.

## 3.8 Summary

The lecture has now assembled the basic language needed for the two-state quantum system. States are normalized vectors in a complex vector space. Observables are represented by Hermitian matrices, whose expectation values and eigenvalues are real. The eigenvalues are the possible measurement outcomes, the corresponding eigenvectors are the states of certainty for those outcomes, and eigenvectors with different eigenvalues are orthogonal. The probability of obtaining a given outcome is the absolute square of the relevant inner product. For spin, the Pauli matrices realize this entire structure concretely, and even along an arbitrary spatial axis the only possible outcomes remain $\pm 1$. What varies from axis to axis is not the list of answers but the probabilities with which those answers occur.